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Valuation of real options using Monte Carlo simulation

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Abstract

This theoretical paper presents a real options valuation method (SCV) based on Wiśniewski's concept, double Monte Carlo simulation (2MC). In this methodology, option valuation involves determining the difference between the average value of a project with an option and the average value of a project without option and is calculated using financial simulation models characterizing these investments. Calculated values of project with option and without depends on the input parameters (deterministic or/and stochastic), that are already used by managers. The real options valuation methodology is presented using a numerical example of a flexible investment. The analysis demonstrates the usefulness of the SCV method, especially in the early stages of investment implementation, for complex and multi-stage projects, and in the presence of multiple sources of risk, that can be easily modeled using market data. An additional conclusion from the paper is the difference between the values of real options, determined using the SCV method, and the value of decision flexibility (2MC method), as well as their economic significance and their application in practice.

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Keywords: Real options, Monte Carlo Simulation, investment appraisal

1. Introduction

1.1. Real options valuation methods

Real options are rarely used in assessing the effectiveness of investments and managing corporate finances. In a study conducted in 2015, it turned out that out of 1,500 CFOs from Denmark, Norway and Sweden, only 6% of

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respondents actively use real options, 70% of CFOs do not know the methodology. For the respondents familiar with the framework, the complexity of real options is the main hinder for implementation [1]. Other studies confirm the small percentage of managers using this concept. For example, among Fortune's 1,000 largest companies, out of 279 respondents, 40 used real options in 2007 (14.3%) [2]. Despite the methodological difficulties described below, real options are sometimes used in various spheres of the economy, e.g. in the energy sector [3], especially renewable [4], [5] and decision making process [6], R&D research (most appropriate in managing technical risk, market risk and technology deployment) and others [7]. Real options are predominantly utilized in business areas with high uncertainty, where management, consciously or unconsciously, regularly creates and exercises these options. In R&D, real options are evident as management retains the right, but not the obligation, to commercialize outcomes. They are relevant for decisions made under conditions of risk and uncertainty, particularly those involving multiple risks and numerous configurations. Real options are also pertinent in scenarios with cascading risks and their associated decisions. The need for methods determining the value of real options and decision flexibility is therefore obvious. What hinders them are methodological issues, in particular complicated valuation algorithms, often based on financial option valuation methods that do not reflect the unique specificity of the investment. Methodological problems translate into relatively little practical application. This happens in particular due to the assumptions that are difficult to meet and the complicated estimation of the input parameters of the valuation model. Scientists are therefore looking for new, simpler methods of valuing real options, e.g. the Datar-Mathews method described below or other methods based on simulation and decision trees [8], [9]. All these attempts are intended to simplify the theory and make it easier to apply. One of the attempts to facilitate the use of real options in practice is the SCV method proposed by the author and described in the methodological part.

The estimation of real options value commonly employs techniques used in financial options valuation. The authors of the most popular valuation model were Nobel laureates Fischer Black and Myron Scholes [10]. It was they who first noticed that a risk-free portfolio (arbitrage portfolio) can be constructed from stocks and options. Since arbitrage ensures that the return on a risk-free portfolio must be equal to the risk-free rate of return, thus, in combination with appropriate boundary conditions, this allowed to derive an analytical model for the valuation of European call options as a function of the stock price, the exercise price, the time to option expiration, the risk-free interest rate, and the variance of stock price. In order for the model to operate efficiently in the financial market, it had to adopt numerous assumptions aimed at creating conditions for a perfect market. It was assumed that: the risk-free rate remains constant throughout the option's expiration period, stock returns over infinitely small time intervals follow a normal distribution, the distribution of stock prices is lognormal, the stock does not pay dividends, short selling is possible, and the movement of stock prices can be modeled by a continuous stochastic process, such as the Ito process. These difficult to meet assumptions (as well as many others not mentioned here) made it possible to determine the analytical formula for the valuation of European call and put options. The Black-Scholes model has had a significant impact on the theory of derivative instrument valuation, market behaviors, and has contributed to the dynamic development of options markets. The Black-Scholes model is a good solution for European call and put options, but its usefulness for pricing more complex, real options is limited. Real options are often American options, they appear as a combination of several simple options and depend on several sources of risk that change over time. Each complicated case of real options would have to have a separate analytical solution, which could be problematic. An exemplary modification of the Black-Scholes formula is its generalization for stocks paying dividends over the life of the option, developed by Robert Merton [11]. An additional parameter y determines the value of the dividend paid or other cash flows in relation to the value of the underlying instrument. In the context of real options, the y parameter is interpreted in various ways: 1) as a reduction in the value of the investment project through the payment of dividend, 2) costs of maintaining the option - license fees, lease costs, 3) the impact of market competition. Despite the difficulties in estimating this parameter, it allows for a simplified consideration of additional factors affecting the value of the option. Analytical methods of valuing real options, i.e. the Black-Scholes model and its extensions (e.g. the above-mentioned Black-Scholes-Merton model), are problematic to apply in practice due to their rigid assumptions. Estimating the value of input parameters (e.g. volatility) to the valuation model is detached from market realities and not very intuitive for managers [12].

Analysts deal with these problems partially by using methods based on the binomial tree, which is a more flexible tool, to value real options. The concept developed by J. Cox, S. Ross and M. Rubinstein [13], as an alternative method to the Black-Scholes formula is an approximation of solutions to continuous-time models [14]. The essence of this method is to build a tree of the values of the underlying instrument, determining the boundary conditions, which leads

to the creation of a tree with option values. There are two main options valuation methods in the binomial tree - the option equivalent method and the risk indifference method [15]. The option value is calculated at the last nodes of the binomial tree, based on boundary conditions and using backward induction. The use of decision trees enables the valuation of complex, multi-stage options. It also allows you to sum up the economic effects of several different options or apply more complex conditions for the exercise of real options. However, it does not solve practical problems related to estimating the value of the underlying asset and its volatility, or taking into account many risk factors. These remain out of reach for many managers, making the use of real options theory in practice quite rare.

An attempt to solve the problems related to the use of the Black-Scholes model and binomial trees is to use fuzzy sets [16] and Monte Carlo simulations for real options valuation. In most cases, the use of Monte Carlo simulations for the valuation of real options involves reproducing price changes of the underlying asset (geometric Brownian motions with a logarithmic-normal distribution) using a stochastic process [17]. The option value is determined based on the average obtained by comparing the value of the underlying instrument and the boundary conditions for each scenario [18].

Another interesting technique for valuing real options, based on Monte Carlo simulation and at the same time consistent with the Black-Scholes model [19], is the Datar-Mathews method [20]. It involves generating a sufficiently large number of scenarios of future operating profits (EBIT, S_T) and costs related to the implementation of the investment. The level of operating profit and investment outlays is treated as a random variable with a specific distribution. Costs are brought to present value using the risk-free rate (r), while operating profits are discounted in each period (t) at a cost of capital reflecting their risk (μ). Then, in each iteration, the value of investment outlays X_T – investment expenditures, discounted at the risk-free rate r) is subtracted from the operating profit, creating a distribution of the value of benefits resulting from the investment. Real option is the expected value (mean, E) from this distribution at time T = 0. The Datar-Mathews method for valuing real options (C) can be written as follows:

$$C = E[max(S_T e^{-\mu t} - X_T e^{-rt}),0] \quad (1)$$

The Datar Mathews method has an advantage over traditional real option pricing based on the Black-Scholes model because it does not rely on the value of the standard deviation (or variance) of parameters as a measure of project uncertainty and it is possible to use various types of density functions to generate distributions of project revenues and costs. This method is highly intuitive and easy to implement in a financial model using Monte Carlo simulation. Moreover, what is worth emphasizing, the valuation is carried out not in the risk-free world, but in the real world, which enables effective risk analysis.

1.2. Simulation comparative valuation

The Simulation comparative valuation (SCV) method is an extension of the concept of real options valuation called double Monte Carlo (2MC) simulation, formulated by Wiśniewski in 2008 [21]. 2MC used Monte Carlo simulation to determine the economic result related to the use (or not) of the flexibility contained in investment projects. This method treats the cash flows of an investment project as the underlying asset, so it facilitates consolidation by aggregating all risk factors based on the MAD method introduced by Copeland and Antikarov [22].

SCV is using commonly known methodological apparatus for assessing investment performance, such as net present value (NPV). Therefore, it is necessary that the SCV approach is consistent with the basic assumptions of the NPV method. Using decision tree analysis and simulation techniques, it becomes possible to determine the value of a real option by extending the analysis of the NPV financial model. SCV is a method of valuing real options that involves a conditional comparison of the expected value of the project in which the real option is included and can be exercised (NPV_{EXT}) and the value of the investment on which the option is built (NPV_{BASE}) according to the following formula:

$$ROV = E[MAX(NPV_{EXT} - NPV_{BASE}; 0)] \quad (2)$$

In the SCV method, the key to determining the value of a real option is to create a simulation financial model of the underlying investment (NPV_{BASE}). A base investment is defined as an investment project that does not take into

account the possibility or impact of exercising a real option. The financial model characterizes the planned investment project by describing the accompanying assumptions e.g. time horizon, cost of capital and basic data on revenues and operating costs and taking into account these variables, their interconnections (in the form of correlation and autocorrelation) and mechanisms influencing the values. Taking into account the most important risk-bearing variables for the value of the project, such as the volume of demand, production, sales, level of prices and costs, CAPEX, and their stochastic description allows in the simulation process to integrate all of them into one, easy-to-interpret distribution, which facilitates the assessment of the value and risk of the project. The value of the extended project (NPV_{EXT}), unlike that obtained from the base investment (NPV_{BASE}), takes into account all expenditure necessary to maintain and execute the real option, as well as the expected benefits accompanying its existence.

The extended financial model enables easy calculation of cash flows related to the exercise of options. It takes into account all direct effects (e.g. increased sales revenues, reduction of production costs, etc.) as well as indirect effects, e.g. tax benefits and synergies. The use of financial modelling allows to precisely determine the difference between the base project and the extended project in which the option is included. Determining the value of a real option in the SCV method involves calculating the expected value from the set of differences between the value of the extended and the base project, maintaining the condition that the value of the option in each iteration must be greater than or equal to zero.

The SCV algorithm is so universal that it can be used to determine the value of any real option, as well as complex options. Nevertheless, the basic and extended models should be individually constructed, adapting their structure and functions to the operational activities and characteristics of the options. There is no need to distinguish between types of options because each type of real option can be individually and precisely reproduced in the financial model. Financial modeling in the SCV method allows for determining the optimal timing of option exercise based on various criteria, and it is particularly useful for complex, multi-stage projects with multiple sources of risk. SCV is flexible and can handle interdependent, sequential, or mutually exclusive options. The method assumes valuation in an incomplete market, where risk is reflected in cash flows and the cost of capital. It enables a comprehensive risk analysis by incorporating stochastic input variables and their interrelationships. The value of the option is determined by comparing simulated financial models that represent scenarios with and without the real option. SCV can also provide additional insights, such as the distribution of the base model and the decision variable, as well as the probability of exercising the option. The SCV method can be practically applied in various investment scenarios characterized by high uncertainty and multiple sources of risk, for example:

- Development of new technologies, to evaluate investment options, where decisions on project continuation are based on the outcomes of initial research and development stages.
- Exploration of natural resources, to assess options related to exploring new deposits of natural resources, where further investment decisions depend on the results of initial drilling and geological surveys.
- Infrastructure expansion, to evaluate options related to expanding existing infrastructure, such as building new facilities or modernizing old ones, where subsequent project stages are carried out based on preliminary results and market analyses.
- Market entry, where decisions on further investments are made based on initial market research and pilot operation outcomes.
- Product development, to evaluate options related to launching new products, where decisions on continuing or halting production depend on market test results and demand analyses.
- Mergers and acquisitions, to analyse options related to strategic decisions, where subsequent steps are based on preliminary due diligence results and financial analyses.

2. Empirical example

An empirical example is provided to demonstrate the capabilities of the SCV method. The example is purely theoretical and the variables were deliberately selected in such a way as to highlight the advantages and disadvantages of the presented method. In order to illustrate the use of the SCV methodology, the growth option was chosen, which gives the opportunity to implement subsequent projects based on the opportunities created by the original project (project with a growth option). It was assumed that a pilot project would be carried out with investment outlays of EUR 100 million, the success of which, estimated at 60% probability, would result in the exercise of the option. The

implementation of the option will consist in initiating a project to build and operate a production plant, the CAPEX of which will amount to EUR 1,000 million. The extension assumes investment in 2024 and five subsequent years of operational activity, ending with the sale of the plant. The main sources of risk are:

- The unit sales price of the product - was described by a normal distribution, in each period the average price is EUR 10 million and the standard deviation is 10%.
- CAPEX - investment expenditure was characterized by a triangular, symmetrical distribution with the most probable value of EUR 1,000 million, maximum of EUR 1,100 million and minimum of EUR 900 million.
- Risk of a pilot project - even distribution, 60% for success.

The remaining assumptions are as follows:

- Annual production, equal to sales volume (there is no inventory), is equal to 100 units and is constant over time.
- The unit production cost is EUR 5 million and is constant over time.
- Fixed costs amount to EUR 600 million and do not change during the operational period.
- Depreciation was specified as 10% per annum of the CAPEX value
- Net working capital - its impact on the option value was omitted from the analysis.
- Income tax is 19%.
- At the end of the fifth and last year of operating activity, the business is sold for a net amount equal to 20% of the initial expenditure (CAPEX).

The above-mentioned assumptions allow for the creation of a financial simulation model in which the value of the project extension is determined, as illustrated in Table 1. Gray cells indicate the random variables of the model - respectively: 1) price - described by a normal distribution and 2) CAPEX value - triangular distribution. The values in these fields are the average or most likely values of these distributions. The NPV result for extension obtained by substituting these values is positive and amounts to EUR 273 million. It should be emphasized that the extended project will be implemented only if the pilot program is successful, which has a 60% chance of success. In the example under consideration, the base investment (NPV_{BASE}) is a pilot program that is described by a simple random variable (uniform distribution, 60% for success). In business practice, the underlying investment may be expressed in a complex financial model whose value depends on many sources of risk.

Table 1. Financial model of investment.

Year	Unit	2024 0	2025 1	2026 2	2027 3	2028 4	2029 5
Price per unit	m. EUR		10,0	10,0	10,0	10,0	10,0
Quantity	quantity		100	100	100	100	100
Revenue	m. EUR		1 000	1 000	1 000	1 000	1 000
Variable costs per unit	m. EUR		5,0	5,0	5,0	5,0	5,0
Variable costs	m. EUR		600	600	600	600	600
Fixed costs	m. EUR		50	50	50	50	50
Depreciation	m. EUR		100	100	100	100	100
Operation costs	m. EUR		750	750	750	750	750
EBIT	m. EUR		250	250	250	250	250
Income tax (19%)	m. EUR		-48	-48	-48	-48	-48
CAPEX	m. EUR	-1 000					
Depreciation	m. EUR		100	100	100	100	100
Net sale of fixed asset	m. EUR						200
Free cash flow (FCF)	m. EUR	-1 000	303	303	303	303	503
WACC	%	10,0%	10,0%	10,0%	10,0%	10,0%	10,0%
PV (FCF)	m. EUR	-1 000	275	250	227	207	312
NPV	m. EUR	271					

In the base scenario (with average and most probable values), if the pilot project is successful, the economic result related to its implementation and start of production will be EUR 173 million (-EUR 100 million + EUR 273 million). If the pilot program fails, the economic result will be -100 million euros. Taking into account the average and most probable values and the probability of success of the pilot project, it is possible to determine the expected value of the decision-making situation in which the theoretical manager could find himself. If the situation develops well, the project value will be 60%*273 million euros, or 163.8 million euros. In case of failure, it will be 40%*-100 million euros, or -40 million euros. To conclude, the expected value will be EUR 123.8 million (EUR 163.8 million - EUR 40 million).

By performing Monte Carlo simulation on the model presented above, it is possible to determine the real option value based on the condition that the option cannot take negative values. All simulations were performed using 100,000 iteration runs, which gave an average standard error of the option value calculated using the Monte Carlo method of EUR 0.41 million [23].

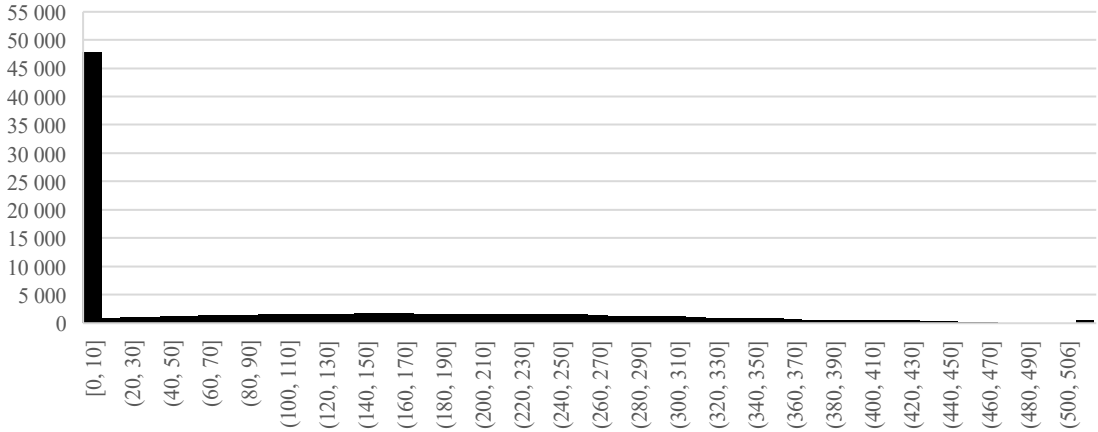

Fig. 1. Distribution of real option value (SCV).

This cuts out all negative scenarios and assigns them a value of zero (see Fig. 1). The average of 100,000 iteration runs is EUR 107 million and the standard deviation is EUR 132 million. Note the high bar near zero (47% of iterations), it aggregates all cases when the pilot project turned out to be a failure and also when the variability of the model's input parameters resulted in negative values after the option was exercised.

The next graph (see Fig. 2) presents the result of a simulation using the same model but showing the economic result associated with the pilot project and the growth option. The difference is that there is no cutoff for negative values - there is no peak around zero (as in Fig. 1). However, there is a high bar around the value of -100 million euros, which means the failure of the pilot project and related losses. In addition, there are other negative values resulting from the variability of the model. In total, as in the case of the option value distribution, negative scenarios (greater than or equal to zero) constitute 47% of all interactions. Since in 40% of cases the extension is not executed, only in about 7% of cases we deal with failed option exercises. This is a situation in which the sum of investment outlays for the pilot project and the NPV of the extended project in a single interaction is less than or equal to zero.

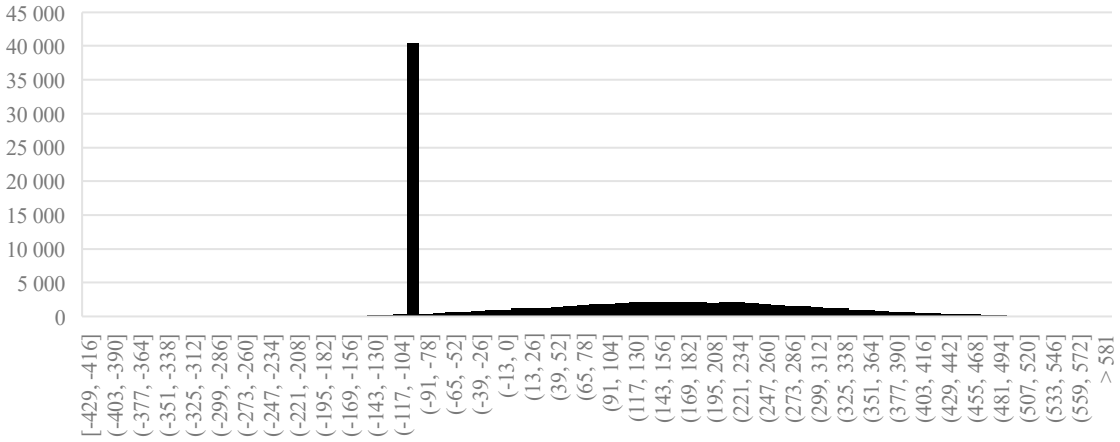

Fig. 2. Distribution of economic result of project with flexibility (2MC).

Therefore, the average value of the economic result of project with flexibility (2MC) is EUR 63 million less than the value of the real option. Additionally, we are dealing with a higher investment risk, as is indicated by the standard deviation of EUR 172 million.

Monte Carlo simulation allows you to determine the sensitivity of option values to individual input parameters, especially those burdened with risk. All analyses are carried out based on reference to the base values of risk-bearing parameters. 100% means CAPEX value of EUR 1,000 million, price of EUR 10 million. The figure below (see Fig. 3) presents a sensitivity analysis of the option value and the NPV of the project with the option due to investment expenditure.

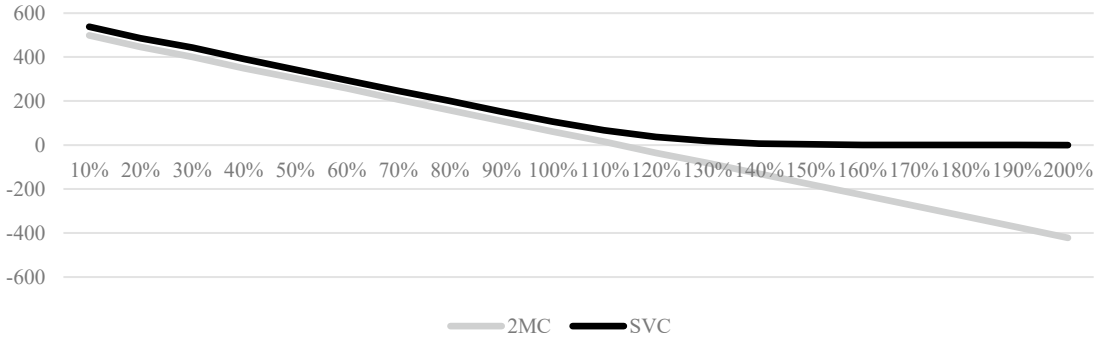

Fig. 3. The distribution of the real option value (SVC) and the distribution of the NPV of the initial investment and option implementation (2MC) depending on the level of CAPEX.

As CAPEX increases, the value of the real option decreases, reaching approximately 130% to zero. The value of decision flexibility due to the lack of value constraints also reaches negative scenarios. When valuing a real option as a possibility, it is assumed that it will be exercised only when it brings benefits. In practice, when planning investment programs, managers should rely on the economic consequences of the decisions made, including negative ones.

The next chart (see Fig. 4) shows the relationship between the real option value (SCV) and elasticity (2MC) depending on the price level. It confirms the observations from the previous chart, when the extended project is ineffective due to low product prices, the option takes the value zero, the result on the implementation of the extended project (2MC) is then negative.

Fig. 4. The distribution of the real option value (SVC) and the distribution of the NPV of the initial investment and option implementation (2MC) depending on the level of prices.

An interesting relationship is shown in the chart below (see Fig. 5). If the price volatility increases from 10% of the standard deviation, the value of the option only increases, taking advantage of the positive deviations of the random variable. In the case of decision flexibility (2MC), when there is no cut-off of negative scenarios, it turns out that implementing an extended project in the event of increased risk is often unprofitable.

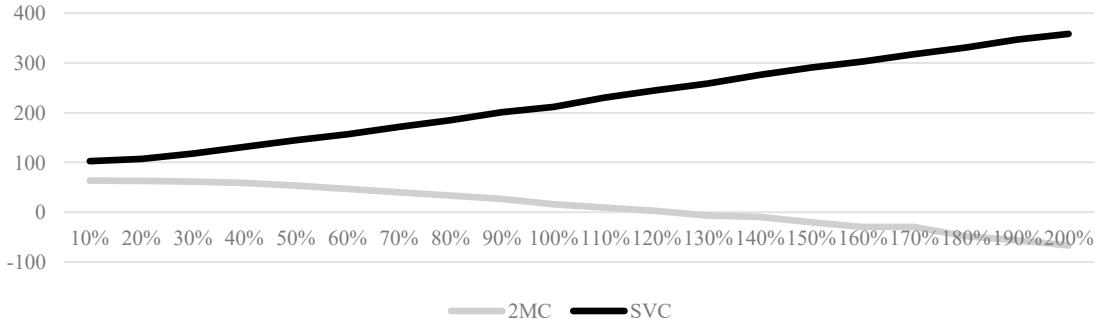

Fig. 5. The distribution of the real option value (SVC) and the distribution of the NPV of the initial investment and option implementation (2MC) depending on the level of volatility.

The next graph (see Fig 6) shows what happens to the option and flexibility values as the probability of pilot project success increases. Activating the option more frequently causes the flexibility value to almost equal the option value at 100% success. The difference results only from bad, negative options execution, i.e. situations in which the extended project is not able to cover the CAPEX expenses for the pilot and extended projects with benefits.

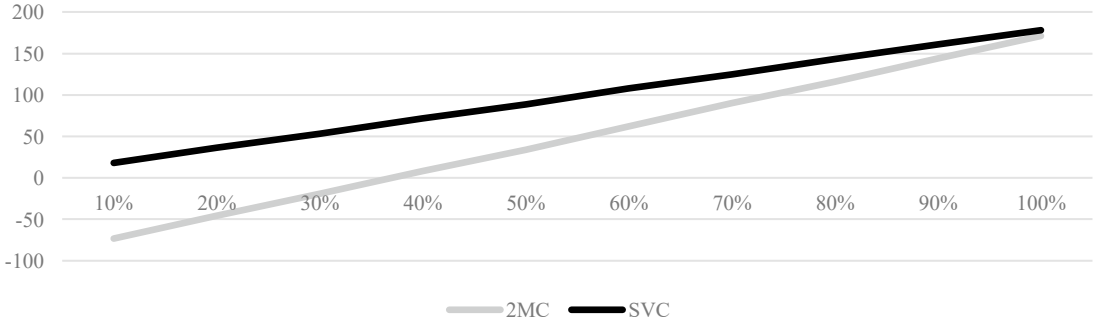

Fig. 6. The distribution of the real option value (SVC) and the distribution of the NPV of the initial investment and option implementation (2MC) depending on the probability of pilot success

To sum up, the differences between the values of real options (SCV) and decision flexibility (2MC) result mainly from their characteristics. Options, as possibilities, cannot achieve negative values. Their potential implementation is perceived as a benefit or, alternatively, in the event of a negative scenario, the option or pilot project is not implemented. It is therefore assumed that the value of real options does not express the result on the implementation of individual scenarios in the way that managers do. There are no wrong executions of options. Therefore, the value of decision-making flexibility (2MC) will probably be closer to practice, as it also takes into account situations in which a decision made on the basis of positive premises (success of the pilot) may turn out to be wrong (negative value of the extended project). The key assumption concerns knowing the value of the project at the time of option exercise. In practice, this value is subject to stochastic fluctuations, which makes the decision to exercise the option burdened with probabilistic considerations. Managers lack knowledge about the strike price of an option due to the inherent uncertainty in its value.

3. Conclusion

The results of the valuation indicate a positive value of the real option (EUR 132 million). The valuation was made by comparing two distributions - a project with and without options, which were exposed to the same risks, i.e. price - normal distribution, CAPEX - triangular distribution and the risk of failure of the pilot project. The analysis showed that the model correctly reproduced the option value depending on individual parameters - sensitivity analysis due to the value of investment outlays, the level of volatility, and prices. For comparison, the valuation of decisional flexibility was also performed using the 2MC method - the result was EUR 63 million, with a higher standard deviation. The 2MC method, unlike real option valuation (SCV), allows for negative scenarios in which the option is exercised, resulting in a negative economic result on the project - hence the lower expected value of decision flexibility.

If the valuation were to be carried out using alternative methods, e.g. the most popular Black-Scholes method, it would be necessary to analyze many theoretical assumptions, the most important of which are:

- The need to identify the underlying asset, the necessity to create a hedging portfolio to mitigate option risk; the presence of a single source of uncertainty, the use of geometric Brownian motion with a constant volatility level to describe changes in the value of the underlying asset. All these assumptions force us to make key decisions related to what the underlying asset is, how its values change, whether the risk is to be constant over time and how to reflect it. An attempt to solve this problem may be the MAD approach, which assumes that the twin (base) asset is the value of the project on which the real option is built. It is estimated using discount methods and usually depends on many risky factors.
- The known and constant strike price of the option - to avoid this issue, decision trees or an appropriate modification of the differential equation should be used.

The indicated assumptions indicate that the Black-Scholes model, used to value options that are not standard options of a European nature, will have to be modified. Estimating volatility or the problem of mapping the underlying asset will require the use of market data and many elements of financial modeling. The SCV and 2MC methods, using the well-known framework of discount valuation methods, bypass many of these assumptions, and additionally use directly market data known to managers.

The SCV method used in this paper has many advantages. It allows you to answer the question not only about the value of the option, but also to determine the optimal date for its exercise, to assess its financial effects and to assess the sensitivity of the option value to stochastic data and exercise conditions. Financial modeling enables its use in the case of complex, multi-stage options with many sources of risk. A very important feature of this method is the ability to estimate input parameters based on market data, which makes it easier for managers to understand and use the method. Despite its advantages, the SCV method is not without disadvantages, in particular the need to use complex financial modeling and the need to perform numerical operations using specialized software such as Crystal Ball or Pallisade.

The SCV method, should be used at an early stage of investment implementation, during capital budgeting, when the possibilities of a given project are checked. However, in investment management, the economic result of realizing flexibility depending on risky decision-making situations will be more useful. The results obtained using the twin 2MC method will therefore be more useful for managers who will certainly appreciate the analysis of conditional decisions made automatically by the model. It is particularly important in risk analysis and management to recognize not only possibilities and options but also negative scenarios.

References

[1] Anders, Horn, Frode, Kjærland, Peter, Molnár, Beate W., & Steen, Beate W. (2015). The Use of Real Options Theory in Scandinavia's Largest Companies. *International Review of Financial Analysis*, 41, 74-81. [Online]. Available: https://doi.org/10.1016/j.irfa.2015.05.026.

[2] Block, Stanley. (2007). Are real options actually used in the real world? *The Engineering Economist*, 52(3), 255-267.

[3] Nadarajah, Selvaprabu, & Secomandi, Nicola. (2022). A Review of the Operations Literature on Real Options in Energy. [Online]. Available: https://ssrn.com/abstract=3877512 or http://dx.doi.org/10.2139/ssrn.3877512.

[4] Sperstad, Iver Bakken, Rana, Rubi, & Sandell, Susanne. (2022). Methodology for Evaluating Grid Development Strategies Considering Real Options and Risks. [Online]. Available: https://doi.org/10.36227/techrxiv.21731489.v1.

[5] Selvaprabu Nadarajah, Secomandi, Nicola. (2023). A review of the operations literature on real options in energy. European Journal of Operational Research, 309(2), 469-487. [Online]. Available: https://doi.org/10.1016/j.ejor.2022.09.014.

[6] Benitez, Guilherme, & Ferreira Lima, Mateus. (2019). The real options method applied to decision making – an investment analysis. Brazilian Journal of Operations & Production Management, 16, 562-571. [Online]. Available: https://doi.org/10.14488/BJOPM.2019.v16.n4.a2.

[7] Chi, Tailan, Li, Jing, Trigeorgis, Lenos, & Tsekrekos, Andrianos. (2019). Real options theory in international business. Journal of International Business Studies, 50. [Online]. Available: https://doi.org/10.1057/s41267-019-00222-y.

[8] Alexander, Carol, & Chen, Xi. (2021). Model risk in real option valuation. Annals of Operations Research, 299, 1025–1056. [Online]. Available: https://doi.org/10.1007/s10479-019-03273-4.

[9] Locatelli, Giorgio, Mancini, Mauro, & Lotti, Giovanni. (2020). A simple-to-implement real options method for the energy sector. Energy, 197, 117226. [Online]. Available: https://doi.org/10.1016/j.energy.2020.117226.

[10] Black, Fisher, & Scholes Myron. (1973). The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3).

[11] Merton, Robert. (1973). The Theory of Rational Option Pricing. *Bell J Econ Manage Sci*, 4, 141-183. https://doi.org/10.1142/9789812701022_0008.

[12] Pawlak, Marcin. (2021). The application of Double Monte Carlo Simulation (2MC) in the valuation of real options. *Scientific Publishing House Centre of Sociological Research*, 82-87. [Online]. Available: DOI: 10.14254/978-83-963452-4-0/2021.

[13] Cox, John, Ross, Steven, & Rubinstein, Mark. (1979). Option Pricing: A Simplified Approach. *Journal of Financial Economics*, 7(3), 229–263.

[14] Brandão, Luiz, Dyer, James, & Hahn, Warren. (2005). Using Binomial Decision Trees to Solve Real-Option Valuation Problems. *Decision Analysis*, 2, 69-88. https://doi.org/10.1287/deca.1050.0040.

[15] Pera, Krystian. (2010). Modyfikacja metody ekwiwalentu pewności dla projektów inwestycyjnych o znanym poziomie wartości strategicznej. *Prace i Materiały Wydziału Zarządzania Uniwersytetu Gdańskiego*, 151–163.

[16] Collan, Mikael, Fullér, Róbert, & Mezei, József. (2009). A Fuzzy Pay-Off Method for Real Option Valuation. *JAMDS*. https://doi.org/10.1109/BIFE.2009.47.

[17] Boyle, Phelim. (1977). Options: A Monte Carlo Approach. *Journal of Financial Economics*, 4, 323–338.

[18] Ziętek-Kwaśniewska, Katarzyna. (2006). Symulacje Monte Carlo jako metoda wyceny opcji. *Scientific Bulletin of Chełm, Section of Mathematics and Computer Science*, (2).

[19] Datar, Vinay, & Mathews, Scott. (2005). European Real Options: An Intuitive Algorithm for the Black-Scholes Formula. *Journal of Applied Finance*, 14, 45–51.

[20] Mathews, Scott, Datar, Vinay, & Johnson, Blake. (2007). A Practical Method for Valuing Real Options: The Boeing Approach. *Journal of Applied Corporate Finance*, 19, 95-104. https://doi.org/10.1111/j.1745-6622.2007.00140.x.

[21] Wiśniewski, Tomasz. (2008). Evaluation of investment effectiveness with particular emphasis on risk. *Szczecin: Scientific Publishing House of the University of Szczecin*.

[22] Copeland, Thomas, & Antikarov, Vladimir. (2003). *Real Options: A Practitioners Guide*. New York: Texere, 244-253.

[23] Geske, Robert, & Shastri, Kuldeep. (1985). Valuation by Approximation: A Comparison of Alternative Option Valuation Techniques. *Financial and Qualitative Analysis*, 20, 45–71.